MIL-G-14562D
29 April 1985
SUPERSEDING
MIL-G-14562C
11 May 1973

MILITARY SPECIFICATION

GUN, FLUSHING, RADIATOR AND ENGINE BLOCK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of gun, which flushes radiators and engine blocks by using compressed air and water under pressure.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206	-	General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	-	Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete gun for first article inspection (see 4.3 and 6.2.1). The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The gun shall be new and of the manufacturer's current design meeting the performance requirements specified herein. The gun shall be designed to operate with a water pressure of 15 to 45 pound-force per square inch gage (psig) and a compressed air pressure range of 60 to 120 psig. The flushing gun shall also be designed to operate simultaneously with air and water at a rate of flow of no less than three gallons per minute (gpm). The design shall be such that the operator can obtain a positive, secure hand grip during gun operation. Those parts of the gun subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the gun or any component.

* 3.2.1 Safety and health requirements. The flushing gun shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the gun itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.4 Construction. The gun shall be furnished complete so it can be used for the operations specified herein. Parts used in fabricating the gun shall be new and free of defects. All part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

* 3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.4.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.4.4 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.4.5 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.6 Surfaces. Surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

* 3.4.7 Protective finishes. All exposed metal (see 6.5.1) and working surfaces, except those made of corrosion-resistant materials, shall be coated or covered with a protective finish that provides protection against corrosion.

3.5 Performance and product characteristics. The gun shall be capable of flushing water, with air pressure, through the radiator or engine block, and heater when applicable. By regulation of the control valves, the gun shall be capable of mixing air and water in any proportion desired by the operator. The gun shall function efficiently when operating from an air source of 60 to 120 psig and a water source of 15 to 45 psig with a rate of flow of no less than three gpm.

3.6 Components.

3.6.1 Body. The body shall be a corrosive-resistant metal casting. A hand grip shall be provided to afford a positive hold by the operator. With air and water valves in the OFF position, the gun shall be capable of withstanding 200 psig air pressure and 75 psig water pressure for a minimum of 60 seconds without leakage or damage to the gun.

3.6.1.1 Air and water connecting threads. The gun shall contain a standard 0.25 - 18 NPT female thread for air hose attachment and a standard 0.75 - 11 1/2 NH female thread for water hose attachment (see 3.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

3.6.2 Adapters.

* 3.6.2.1 Radiator and block hose adapters. The radiator and block adapter, or adapters, shall be of the graduated or self-holding type and shall accommodate radiator and engine block hoses having a minimum inside diameter of no more than 1.000 inch up to, and including, a maximum inside diameter of no less than 2.000 inches. The adapters shall approximate a cone in shape and may be an integral part of the gun body or be detachable.

* 3.6.2.2 Heater hose adapter. Unless otherwise specified (see 6.2.1), a heater hose adapter shall be provided with the gun and may be an integral part of the gun body or be detachable.

* 3.6.3 Air control valve. An air control valve shall be provided which can be operated independently of, or simultaneously with, the water control valve. The valve may be a button, trigger, or lever type.

* 3.6.4 Water control valve. A water control valve shall be provided which can be operated independently of, or simultaneously with, the air control valve. The valve may be a button, trigger, or lever type.

* 3.6.5 Discharge hose assembly. Unless otherwise specified (see 6.2.1), a discharge hose assembly 30 inches ± one inch in length shall be provided to facilitate flushing when space is limited or hose connections which are not readily accessible are encountered. The hose shall have a connection on one end to fit the gun discharge outlet or adapters, and a means for attaching to the radiator, engine block, or heater on the other end.

3.7 Product identification. The gun shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.8 Workmanship. Standards of workmanship shall assure that the gun shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:26Z
Check the source to verify that this is the currentversion before use.

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Pressure test (see 4.6.1).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Examination	General level II	1.0
Pressure test	General level II	1.0
Packaging inspection	Special level S-4	4.0

* 4.5 Examination. Visually and manually examine the gun to determine conformance with the requirements of 3.2 through 3.4.7 and 3.6.1 through 3.8. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. A certificate of conformance may be used to satisfy examination requirements when more than a visual or dimensional verification would be required. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the gun and its components have passed the examination prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

* 4.6.1 Pressure test. The gun shall be connected to an air supply pressure of 180 to 200 psig and to a water supply with a pressure of 70 to 75 psig. With pressures off, the air and the water control valves shall be operated at least once through the full range and then set in the OFF position. Full air and water pressure shall then be applied to the gun for 60 seconds and the gun tested for air and water leaks by immersion or application of soap solution. Failure of the gun to pass the test, as evidenced by any air or soap bubbles or by damage to the gun, shall be cause for rejection (see 3.6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

4.6.2 Operational test. The gun shall be connected to an operating air supply of 60 to 120 psig and a water supply line of 15 to 45 psig. The gun shall then be used to reverse flush radiators, engine blocks, and hot water heaters. Two different types of vehicles shall be used having radiator or hose diameters of the minimum and maximum specified in 3.6.2.1. The ease of air and water valve operation, adjustment, and metering shall be checked for compliance with the requirements specified in 3.5, 3.6.3, and 3.6.4.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The gun and accessories shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Preservative application. Interior surfaces of the gun and adapters shall be coated with a preservative oil conforming to type P-10, grade 30, of MIL-P-116.

* 5.1.1.3 Unit packaging. The unit method of packaging shall conform to method IC-2 of MIL-P-116. Apertures of the gun shall be sealed with plastic caps or plugs conforming to MIL-C-5501 or with tape conforming to PPP-T-60, type IV. The gun and adapters shall be individually wrapped with barrier material conforming to MIL-B-121, type II, grade A. When applicable, the discharge hose shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling. Technical publications, when applicable, shall be packaged in accordance with method IC-1 of MIL-P-116. The unit container for the gun, accessories, and publications shall conform to PPP-B-676, variety optional.

* 5.1.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.2.1 Level A. Guns preserved and packaged as specified in 5.1.1 shall be packed in an exterior shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of each shipping container shall not exceed the weight limitations of the applicable box specification. As far as practicable, containers shall be uniform in size and contain like quantities. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Container closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. The guns shall be packed as specified in 5.2.1 except the shipping container shall conform to PPP-B-640, class 2.

* 5.2.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

* 5.3 Marking. Marking shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military (Levels A and B) marking is required, it shall be so specified in the contract.

* 5.3.1 Level A and B marking. Level A and B marking shall be as specified in MIL-STD-129.

* 5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. This specification covers a flushing gun intended for reverse flushing of vehicle radiators, engine blocks, and hot water heaters. This flushing gun uses a combination of water and air, regulated by control valves, to produce a steady pressure and flow, or a pulsating pressure and flow, to effect the removal of scale and dirt and to eliminate clogging caused by foreign particles or the accumulation of scale. Through the use of hose adapters, any size cooling system may be flushed.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.1 and 6.4).
(d) When heater hose adapter not required (see 3.6.2.2).
(e) When discharge hose is not required or varying dimensions are required (see 3.6.5).
(f) Product identification if different (see 3.7).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling if different (see 4.4.1).
(j) Level of preservation, packing and marking if different (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 6.3.1).

* 6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

* 6.5 Definitions.

6.5.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, solvents, and flushing and cleaning chemicals and fluids (see 3.4.7).

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Air Force - 82
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - AL

Project No. 4910-0538

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.

reduced to the simplest form of procedures, requiring a minimum of skills and

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Materiel Readiness
Command
Attn: DRSAR-LEE-S
Rock Island, IL 61299

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.